

- Podemos representar una base mediante el funcional.

6. En el caso tridimensional tenemos que, si  $\{e_i\}$  define un sistema de coordenadas (dextrogiro) y no necesariamente ortogonal, demuestre que:

a)

$$e^i = \frac{e_j \times e_k}{e_i \cdot (e_j \times e_k)}, \quad i, j, k = 1, 2, 3 \text{ y sus}$$

permutaciones cíclicas.

$$\bullet \quad e^1 = \frac{e_2 \times e_3}{e_1 \cdot (e_2 \times e_3)} = \frac{e^1}{e_1 \cdot e^1} = \frac{e^1}{1} = e^1,$$

$$\bullet \quad e^3 = \frac{e_1 \times e_2}{e_3 \cdot (e_1 \times e_2)} = \frac{e^3}{e_3 \cdot e^3} = \frac{e^3}{1} = e^3,$$

$$\bullet \quad e^2 = \frac{e_3 \times e_1}{e_2 \cdot (e_3 \times e_1)} = \frac{e^2}{e_2 \cdot e^2} = \frac{e^2}{1} = e^2.$$



b) Si los volúmenes:  $V = e_1 \cdot (e_2 \times e_3)$  y  $\tilde{V} = e^1 \cdot (e^2 \times e^3)$  entonces  $V\tilde{V} = 1$ .

$$V\tilde{V} = [e_1 \cdot (e_2 \times e_3)][e^1 \cdot (e^2 \times e^3)] \\ = [e_i \cdot e^i][e^i \cdot e_i] = 1 \cdot 1 = 1$$

c) ¿Qué vector satisface  $a \cdot e^i = 1$ ? Demuestre que  $a$  es único.

Asumamos que  $a \cdot e^i = 1$ , pero por definición

$e_i \cdot e^i = 1$  también, de este modo  $a = e_i$  y  $a$  es único  
es decir para cada  $e^i$  existe un  $e_i$  tal que  $e^i \cdot e_i = 1$ .

d) Encuentre el producto vectorial de dos vectores  $a$  y  $b$  que están representados en un sistema de coordenadas oblicuo: Dada la base  $w_1 = 4\hat{i} + 2\hat{j} + \hat{k}$ ,  $w_2 = 3\hat{i} + 3\hat{j}$ ,  $w_3 = 2\hat{k}$ . Entonces encuentre:

I. Las bases recíprocas  $\{e^i\}$

II. Las componentes covariantes y contravariantes del vector  $a = \hat{i} + 2\hat{j} + 3\hat{k}$ .

I) Cualquier vector en la base  $w$  se puede escribir como:

$|v\rangle = a_i |w_i\rangle$  y supondremos que la base recíproca satisface la condición:

$\langle e^i | w_j \rangle = \delta_j^i$ . Entonces

$$\langle e^1 | w_1 \rangle = (a_1, b_1, c_1) \frac{1}{\sqrt{21}} \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{21}} (4a_1 + 2b_1 + c_1) = 1$$

$$\langle e^1 | w_2 \rangle = (a_1, b_1, c_1) \frac{1}{\sqrt{18}} \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{18}} (3a_1 + 3b_1) = 0$$

$$\langle e^1 | w_3 \rangle = (a_1, b_1, c_1) 2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 2(c_1) = 0$$



Resolviendo el sistema de ecuaciones

$$\frac{4a_1}{\sqrt{21}} + \frac{2b_1}{\sqrt{21}} + \frac{c_1}{\sqrt{21}} = 1 \Rightarrow \langle e^1 | =$$

$$\frac{3}{\sqrt{18}} a_1 + \frac{3}{\sqrt{18}} b_1 = 0$$

$$2c_1 = 0$$

$$\langle e^1 | = (\sqrt{21}/2, -\sqrt{21}/2, 0)$$

$$\langle e^2 | w_1 \rangle = (a_2, b_2, c_2) \frac{1}{\sqrt{21}} \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{21}} (4a_2 + 2b_2 + c_2) = 0$$

$$\langle e^2 | w_2 \rangle = (a_2, b_2, c_2) \frac{1}{\sqrt{18}} \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{18}} (3a_2 + 3b_2) = 1$$

$$\langle e^2 | w_3 \rangle = (a_2, b_2, c_2) 2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 2c_2 = 0$$

$$\frac{1}{\sqrt{21}} (4a_2 + 2b_2 + c_2) = 0$$

$$\frac{1}{\sqrt{18}} (3a_2 + 3b_2) = 1$$

$$2c_2 = 0$$

$$\langle e^2 | = (-\sqrt{2}, 2\sqrt{2}, 0)$$

$$\langle e^3 | w_1 \rangle = (a_3, b_3, c_3) \frac{1}{\sqrt{21}} \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{21}} (4a_3 + 2b_3 + c_3) = 0$$

$$\langle e^3 | w_2 \rangle = (a_3, b_3, c_3) \frac{1}{\sqrt{18}} \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{18}} (3a_3 + 3b_3) = 0$$

$$\langle e^3 | w_3 \rangle = (a_3, b_3, c_3) 2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 2c_3 = 1$$

$$\langle e^3 | = (-1/4, 1/4, 1/2)$$



II) Debemos escribir el vector  $a$  en términos de las bases  $\{w_i\}$ ,  $\{e^i\}$ , entonces.

$$a = \hat{i} + 2\hat{j} + 3\hat{k} = \alpha w_1 + \beta w_2 + \gamma w_3,$$

$$\Rightarrow \alpha(4\hat{i} + 2\hat{j} + \hat{k}) + \beta(3\hat{i} + 3\hat{j}) + \gamma(2\hat{k})$$

$$\Rightarrow \alpha 4\hat{i} + \alpha 2\hat{j} + \alpha \hat{k} + \beta 3\hat{i} + \beta 3\hat{j} + \gamma 2\hat{k}$$

$$= 4\alpha + 3\beta = 1$$

$$2\alpha + 3\beta = 2$$

$$\alpha + 2\gamma = 3$$

$$\left. \begin{array}{l} 4\alpha + 3\beta = 1 \\ 2\alpha + 3\beta = 2 \\ \alpha + 2\gamma = 3 \end{array} \right\} (\alpha, \beta, \gamma) = (-1/2, 1, 7/4).$$

componentes covariantes.

$$a = \hat{i} + 2\hat{j} + 3\hat{k} = \alpha e^1 + \beta e^2 + \gamma e^3$$

$$\Rightarrow \alpha \left( \frac{\sqrt{21}}{2} \hat{i} - \frac{\sqrt{21}}{2} \hat{j} \right) + \beta (-\sqrt{2} \hat{i} + 2\sqrt{2} \hat{j}) + \gamma \left( -\frac{1}{4} \hat{i} + \frac{1}{4} \hat{j} + \frac{1}{2} \hat{k} \right)$$

$$= \alpha \frac{\sqrt{21}}{2} \hat{i} - \alpha \frac{\sqrt{21}}{2} \hat{j} - \beta \sqrt{2} \hat{i} + \beta 2\sqrt{2} \hat{j} - \gamma \frac{1}{4} \hat{i} + \gamma \frac{1}{4} \hat{j} + \gamma \frac{1}{2} \hat{k}.$$

$$= \frac{\sqrt{21}}{2} \alpha - \sqrt{2} \beta - \frac{1}{4} \gamma = 1$$

$$-\frac{\sqrt{21}}{2} \alpha + 2\sqrt{2} \beta + \frac{1}{4} \gamma = 2$$

$$\frac{1}{2} \gamma = 3$$

$$\left. \begin{array}{l} \frac{\sqrt{21}}{2} \alpha - \sqrt{2} \beta - \frac{1}{4} \gamma = 1 \\ -\frac{\sqrt{21}}{2} \alpha + 2\sqrt{2} \beta + \frac{1}{4} \gamma = 2 \\ \frac{1}{2} \gamma = 3 \end{array} \right\} (\alpha, \beta, \gamma) = \left( \frac{11\sqrt{21}}{21}, \frac{3\sqrt{2}}{2}, 6 \right)$$

componentes contravariantes.



Ahora bien, derivando y usando la regla de la cadena

$$x^i = x^i(x^{j'}(x^m)) \Rightarrow \frac{\partial x^i}{\partial x^m} = \frac{\partial x^i}{\partial x^{j'}} \frac{\partial x^{j'}}{\partial x^m} = \delta_m^i$$

$$\Rightarrow dx^i = \frac{\partial x^i}{\partial x^{j'}} dx^{j'}$$

Para transformar las componentes de tensor

$$T_{j_1 \dots j_m}^{i_1 \dots i_n} = \frac{\partial x^{i_1}}{\partial x^{k_1}} \dots \frac{\partial x^{i_n}}{\partial x^{k_n}} \frac{\partial x^{l_1}}{\partial x^{j'_1}} \dots \frac{\partial x^{l_m}}{\partial x^{j'_m}} T_{k_1 \dots k_n}^{l_1 \dots l_m}$$

7. Considere una vez más el espacio vectorial de matrices hermiticas  $2 \times 2$  y la definición de producto interno  $\langle a|b \rangle = \text{Tr}(A^\dagger B)$ . Hemos encontrado que las matrices de Pauli y las unitarias forman base para ese espacio. Encuentre la base dual asociada a las bases de Pauli y, adicionalmente, dado un vector genérico en este espacio vectorial encuentre su 1-forma asociada.

$$\text{Sean } \sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Existe una base en el espacio dual tal que se cumple

$$\langle \sigma_i | z_j \rangle = \delta_j^i$$

$$\langle \sigma_0 | z_0 \rangle = \text{tr} \left[ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \right] = \text{tr} \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix}$$

$$= a_1 + a_4 = 1 \quad \checkmark$$



$$\langle \sigma_0 | z_1 \rangle = \text{tr} \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix} = b_1 + b_4 = 0$$

$$\langle \sigma_0 | z_2 \rangle = \text{tr} \begin{pmatrix} c_1 & c_2 \\ c_3 & c_4 \end{pmatrix} = c_1 + c_4 = 0$$

$$\langle \sigma_0 | z_3 \rangle = \text{tr} \begin{pmatrix} d_1 & d_2 \\ d_3 & d_4 \end{pmatrix} = d_1 + d_4 = 0$$

$$\langle \sigma_1 | z_0 \rangle = \text{tr} \left[ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \right] = \text{tr} \begin{pmatrix} a_3 & a_4 \\ a_1 & a_2 \end{pmatrix} = a_3 + a_2 = 0$$

$$\langle \sigma_1 | z_1 \rangle = \text{tr} \left[ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix} \right] = \text{tr} \begin{pmatrix} b_3 & b_4 \\ b_1 & b_2 \end{pmatrix} = b_3 + b_2 = 1$$

$$\langle \sigma_1 | z_2 \rangle = \text{tr} \begin{pmatrix} c_3 & c_4 \\ c_1 & c_2 \end{pmatrix} = c_3 + c_2 = 0$$

$$\langle \sigma_1 | z_3 \rangle = \text{tr} \begin{pmatrix} d_3 & d_4 \\ d_1 & d_2 \end{pmatrix} = d_3 + d_2 = 0$$

$$\langle \sigma_2 | z_0 \rangle = \text{tr} \left[ \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \right] = \text{tr} \begin{pmatrix} -ia_3 & -ia_4 \\ ia_1 & ia_2 \end{pmatrix} = -ia_3 + ia_2 = 0$$

$$\langle \sigma_2 | z_1 \rangle = \text{tr} \begin{pmatrix} -ib_3 & -ib_4 \\ ib_1 & ib_2 \end{pmatrix} = -ib_3 + ib_2 = 0$$

$$\langle \sigma_2 | z_2 \rangle = \text{tr} \begin{pmatrix} -ic_3 & -ic_4 \\ ic_1 & ic_2 \end{pmatrix} = -ic_3 + ic_2 = 1$$

$$\langle \sigma_2 | z_3 \rangle = \text{tr} \begin{pmatrix} -id_3 & -id_4 \\ id_1 & id_2 \end{pmatrix} = -id_3 + id_2 = 0$$

$$\langle \sigma_3 | z_0 \rangle = \text{tr} \left[ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \right] = \text{tr} \begin{pmatrix} a_1 & a_2 \\ -a_3 & -a_4 \end{pmatrix} = a_1 - a_4 = 0$$

$$\langle \sigma_3 | z_1 \rangle = \text{tr} \begin{pmatrix} b_1 & b_2 \\ -b_3 & -b_4 \end{pmatrix} = b_1 - b_4 = 0$$

$$\langle \sigma_3 | z_2 \rangle = \text{tr} \begin{pmatrix} c_1 & c_2 \\ -c_3 & -c_4 \end{pmatrix} = c_1 - c_4 = 0$$

$$\langle \sigma_3 | z_3 \rangle = \text{tr} \begin{pmatrix} d_1 & d_2 \\ -d_3 & -d_4 \end{pmatrix} = d_1 - d_4 = 1.$$



Resolviendo el sistema de ecuaciones

$$|z_0\rangle = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}, \quad |z_1\rangle = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$$

$$|z_2\rangle = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}, \quad |z_3\rangle = \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix}$$

Sea un vector  $|u\rangle$  en el espacio vectorial,

$$|u\rangle = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} = \alpha \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} + \beta \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$$

$$+ \gamma \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix} + \delta \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix}$$

$$= \begin{pmatrix} \alpha/2 & 0 \\ 0 & \alpha/2 \end{pmatrix} + \begin{pmatrix} 0 & \beta/2 \\ \beta/2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \gamma/2 \\ -\gamma/2 & 0 \end{pmatrix} + \begin{pmatrix} \delta/2 & 0 \\ 0 & -\delta/2 \end{pmatrix}$$

De este modo las componentes de la matriz  $|u\rangle$  se escriben como:

$$a_1 = \frac{\alpha + \delta}{2}, \quad a_2 = \frac{\beta + \gamma}{2}$$

$$a_3 = \frac{\beta - \gamma}{2}, \quad a_4 = \frac{\alpha - \delta}{2}$$

$$\Rightarrow \frac{1}{2} \begin{pmatrix} \alpha + \delta & \beta + \gamma \\ \beta - \gamma & \alpha - \delta \end{pmatrix}$$